Four-Layer ESP32-C3 Embedded Hardware Architecture with Integrated Power-Path Control (Fusion 360)

Objective: To design and implement a four-layer mixed-signal PCB in Autodesk Fusion 360 based on the ESP32-C3-WROOM-02 module, integrating structured power-path battery management, multi-protocol communication, environmental sensing, analog signal acquisition, and expandable storage within a compact embedded hardware architecture.

System Architecture: The system is architected as a 4-layer embedded hardware platform centered around the ESP32-C3-WROOM-02 microcontroller. The design is divided into two primary domains power management architecture and signal processing architecture.

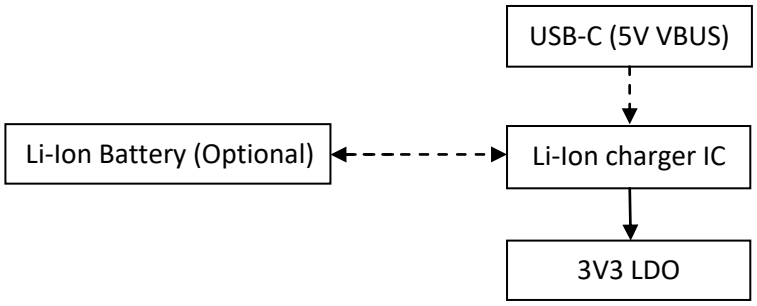

Fig.1: Power Flow Block Diagram

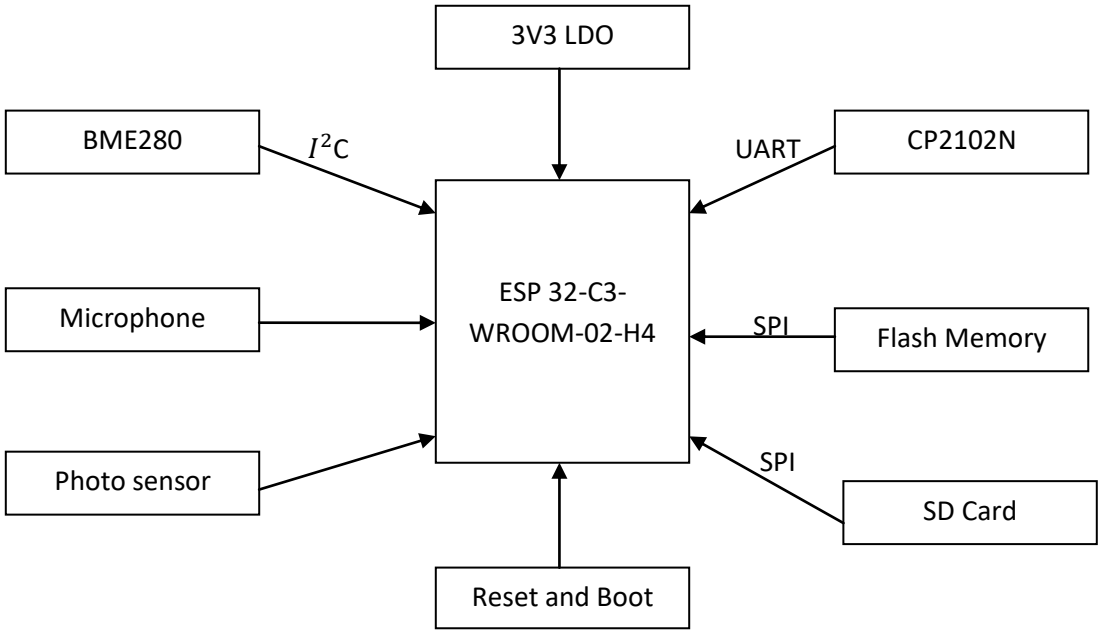

Fig.2: Signal Flow Block Diagram

Power Architecture & Operational Modes: This system runs by 2 modes depending upon the 5V USB connection.

1st mode (**USB connected**) – It is connected to the MCP73871 charger IC it feeds to both Li-Ion charging and also to the remaining components through LM1117 LDO IC by regulating into 3.3V. 2nd mode (**USB not connected**) – Now the MCP73871 charger IC is fed by the charged Li-Ion battery then it feeds to the LM1117 LDO IC by which all components gets the power.

Sensor & Communication Architecture: The ESP32-C3-WROOM-02-H4 operates as the central controller, interfacing with digital and analog peripherals. Sensor acquisition and data exchange are performed using I^2C, SPI, UART, ADC, and GPIO interfaces under master control of the ESP32.

I^2C Interface – BME280 Environmental Sensor: The BME280 is used for environmental monitoring and is operated at 3.3 V within its supported range of 1.71 V to 3.6 V.

- Integrated temperature, humidity, and pressure sensing
- Resolution:
  Temperature: 0.01 °C
  Humidity: 0.008 %RH
  Pressure: 0.16 Pa

It communicates with the ESP32 using the I^2C protocol. The interface uses two lines — **SDA** (data) and **SCL** (clock) — with pull-up resistors to 3.3 V. The ESP32 functions as the I^2C master, initiating communication and reading sensor registers at standard (100 kHz) or fast-mode (400 kHz) speeds.

SPI Interface – Storage Devices: The system incorporates external non-volatile storage through a 32-Mbit W25Q32JV flash memory and an SD card interface for removable data logging. Both devices operate at 3.3 V CMOS logic and are connected to the ESP32 using a shared SPI bus.
The SPI interface uses 4 primary signals
MOSI (Master Out Slave In)
MISO (Master In Slave Out)
SCLK (Serial Clock)
CS (dedicated Chip Select) lines for each device. While MOSI, MISO, and SCLK are common to both storage components, independent CS signals ensure that only one device is active on the bus at a time.
The ESP32 operates as the SPI master, generating the clock and controlling device selection through CS arbitration, enabling reliable firmware storage and data logging over a shared high-speed serial interface.

UART Interface – Programming & Debugging: The ESP32-C3 is connected to a CP2102N USB-to-UART bridge for firmware programming and serial debugging. The CP2102N operates at 3.3 V logic levels and converts USB data from the host into UART signals (TX/RX) compatible with the ESP32. Control lines (DTR/RTS) are utilized for automatic reset and boot mode selection during firmware upload.
Analog & GPIO Interfaces: Analog sensing elements are interfaced to the ESP32-C3 using its internal 12-bit SAR ADC, operating within the 0–3.3 V input range and referenced to the system ground plane for stable conversion.

- **Ambient Light Sensor – TEMT6000:** A phototransistor-based sensor producing current proportional to light intensity. A load resistor converts this into a voltage, which is sampled by the ESP32 ADC via GPIO1. The resulting voltage variation enables quantification of ambient light levels.

- **Microphone Front-End – MAX4466:** Operates from 2.4 V–5.5 V with adjustable gain (25×–125×). The amplified, rail-to-rail output is biased within the ADC range and sampled by the ESP32 through GPIO4.

ADC Characteristics of ESP32-C3: The ESP32 integrates a 12-bit SAR ADC with an input range of 0–3.3 V, providing 4096 discrete quantization levels. The effective voltage resolution is approximately: (3.3V/4096)≈0.8 mV per LSB.

PCB Design & Layout Strategy

The PCB was implemented using a four-layer stack with Layer 2 dedicated to a continuous ground plane to provide controlled return paths and low-impedance reference for signal integrity. Layer 3 was used for structured power distribution with dedicated copper zones for +3.3V, VBUS, and +5V_USB.

The primary system power path from the MCP73871 charger IC (OUT pin) to the LDO regulator was routed using a 1.0 mm trace width to minimize voltage drop and support peak load current. Other supply rails, including VBUS and the regulated 3.3 V domain, were routed at 0.5 mm to balance current handling capability and routing efficiency. Low-current digital and communication signals (SPI, I²C, UART, GPIO) were routed using 0.25 mm traces.

Signal routing was maintained above a continuous ground reference to prevent return path discontinuities. USB differential lines were routed symmetrically, and analog ADC inputs were physically separated from high-speed digital traces to reduce switching noise coupling.

Dedicated test points were added on critical power and communication pins to facilitate probing and system-level debugging.

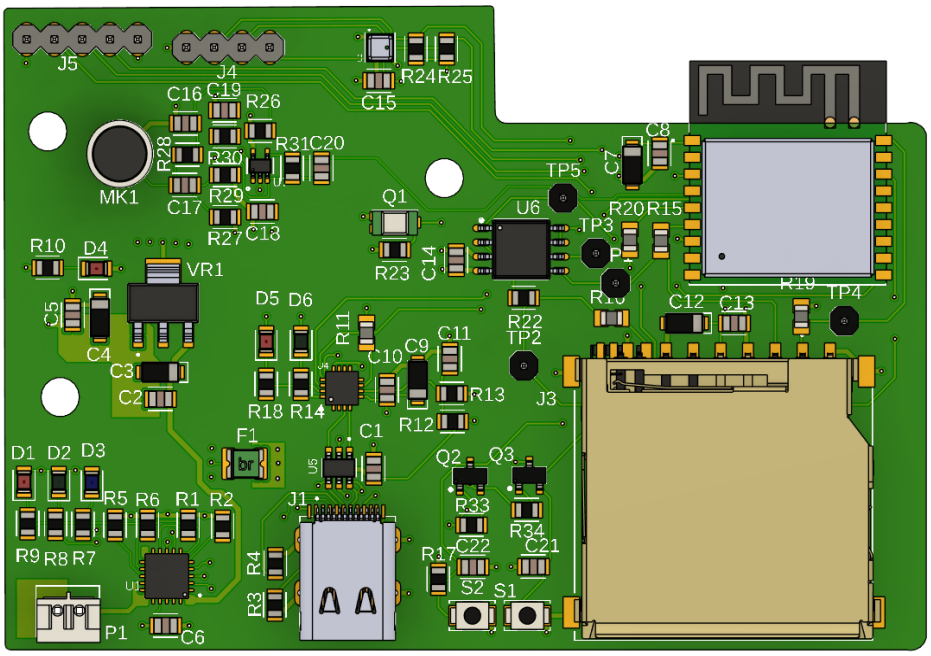

Fig.3: 3D Board View of the Design